
MA 610: MA 610

Course Code	Mathematical Modeling
Course Name	MA 610
L-T-P-C	3-0-0-3
Prerequisites	IC 110, IC 111 for BTech, Ordinary Differential Equations for M.Sc./MTech/PhD
Intended for	Elective for M.Sc./ MTech/PhD/BTech (All Branches)
Mutual Exclusion	NA
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Aim and history, A few simple examples, what is a model, The process of mathematical modeling, Model classification (4 hours)
- **Optimization models:** One variable optimization, some additional materials on population, Multi variable optimization, Computational methods for optimization, some materials on simplex method, Discrete Models - World population growth data snooping, Linear models, Logistic models, Theorems on stability (7 hours)
- **Probability models:** Introduction, Discrete probability models, Continuous probability models, Models for population, Introduction of Population models: Malthus model, Population growth: Logistic model, Harvesting, Population of interacting species: Lotka-Volterra systems (some additional materials), Age-dependent population models. (7 hours)
- **Applications of mathematical modeling:** Mainly mathematical models to study and understand phenomena in chemistry, biology, engineering, political sciences, business and in social sciences. (6 Hours)
- **Laboratory/practical/tutorial Modules:** Individual final project: During the last two weeks of the semester, each student will carry out a project investigating a new mathematical model or carrying out a significant extension of an existing mathematical model discussed in class (6 Hours)

Text Books

1. F. R. Giordano, W. P. Fox, S. B. Horton and M. D. Weir, **A First Course in Mathematical Modeling**, 4th Edition, Brooks/Cole Publishing Company, 2009.
2. **Mathematical Modelling: A Tool for Problem Solving in Engineering, Physical, Biological and Social Sciences**, Pergamon 1990.

References

1. D.N.P. Murthy and N.W. Page and E.Y. Rodin, **Mathematical modeling: a tool for problem solving in engineering, physical, biological, and social sciences**, Pergamon Press, 1990.
2. Dick Clements, **Mathematical modeling: a case study approach**, Cambridge University Press, 1989.
3. T.P. Dreyer, **Modeling with Ordinary Differential Equations**, CRC Press, 1993.
4. Clive L. Dyne, **Principles of Mathematical Modelling**, Academic Press, 2004.
5. M. M. Meerschaert, **Mathematical Modeling**, 2nd Edition, Academic Press, 1998